

# ActInf MathStream 008.2 ~ R Servajean: Intro to Bayesian mechanics: paths-based formalism (part 2)

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hello and welcome everyone it's February 2nd 2024 and we are returning for active math stream 8.2 with Richard seron introduction to basian mechanics free energy principle and the paths based formalism thank you Richard for joining for this part two looking forward to your presentation and discussion I have a weird Echo um wait are you in general synchrony with yourself on YouTube um wait let me see okay perfect sorry um okay great for hi everyone and thank you again um Daniel so today I'm going to present the path-based uh formulation of the frery principle um and I'm going to start by a brief reminder of a couple of things we discussed last time how however if you're not acquainted at all with the Notions of synchronization map of um variational inference or L equation it can be a good idea to uh watch the first video and if you do so please check the comment I posted below or I precise um and correct few things so a quick reminder of where we started and where we so basically if you consider such sparely coupled random dynamical system where  $\mu$  corresponds to the internet States Ita to the external States and A and S the blocket states to respectively the active and sensory States and they form together the So-Cal Mark of blanket uh and just a bit more vocabulary a and  $\mu$  form together the so-called autonomous States and finally if you add the sensory states to the autonomous States so you consider the world Mark of blanquet plus the internal States uh you end up with the so-called particular States  $\pi$  and the particular States defines or constitute a particle an agent bacteria in my schematic for instance so if you consider such sparely C random dynamical system you can interpret the Dynamics of the autonomous States in terms of perceptual and active inference more precisely you can see here in the bottom of my slide that the autonomous mode minimize a free energy function which is defined under a generative model which is typically The Nest density the non equilibrium steady state density uh of your system so more precisely the internal States  $\mu$  parameterize the density  $Q(\mu)$  over external States and if you look at the path of the mode it is so that the the free energy associated with your variational density is always minimize and this license is an

interpretation in terms of Bayesian inference and about action because you can see here that the active uh mode is also a minimizer of the free energy in fact it minimizes the surprise term within free energy so that you can see the particle as actively sampling unsurprising or preferred uh sensory States sensory inputs which is active inference and yeah if you remember well the last video we saw that the generative model encodes the preferences of your system so that was basically where we started and where we ended last time and today I want to uh relax a bit of hypothesis so first of all uh no steady state is assumed we could be at steady States we could have a well defined State density uh but we do know we don't do any assumptions here there is no steady state assumed the second so it extends in a way the scope of the free energy principle the second thing we do is to relax the white noise assumption this means that we don't deal anymore with infinitely rough fluctuations but we have uh uh fluctuations which uh which are smooth uh so basically we deal with colored or correlated noise and this means that the fluctuations become differentiable up to a certain order so basically instead of dealing just with  $x$  and  $\dot{x}$  let's say you have  $\ddot{x}$  and so on and so forth you have higher dynamical orders if you will note that this Vector  $X$  here which precisely corresponds to  $x$   $\dot{x}$  and so on is called the generalized States and the shifted Vector this one where the first component is  $\dot{x}$  we refer to it as the generalized motion and we also say that we work in so-called generalized coordinates of motion not to be confused with generalized coordinates in analytical mechanics so a lot of words but we just we are just saying that the fluctuations are smooth now and we can differentiate uh these fluctuations up to a certain order and this order is the order of generalized motion and it tells you how large the correlation uh length of the fluctuation is basically I also want to introduce the notion of generalized State space so basically you can augment State space with uh generalized motion so instead of just looking at  $x$  the state of your system you look at  $\dot{x}$   $\ddot{x}$  Etc and so basically we consider the so-called generalized State space okay so um a lot once again a lot of words just to say that fluctuations are smooth now so if you look at the equation below it seems that there are dots and primes here which and it seems a bit redundant so in fact what we call  $\dot{x}$  here you can view it as the actual time derivative um that you can evaluate anywhere for instance at  $t$  equal to  $t_0$  and the  $\ddot{x}$  here is just the second component of the momental reg generalized states of your system uh for instance at  $t$  equal to  $t_0$  so that if you evaluate your time derivative at  $t_0$  it equates indeed  $\dot{x}$  but this distinction between the two will become more clear later in the context of generalized filtering so the cool thing here is that if you perform a Taylor expansion of the state of your

system so you have an equation like that you can see that generalized States constitute the coefficient of your expansion indeed you can see in this formula here that you have  $x \cdot x'$   $x'x'$  Etc so basically you do a Taylor expansion around equal to basically and the coefficients of your expansions are generalized States so in a way you can see generalized states are as encoding a path as encoding the future of your uh of the state of your system thanks to this Taylor expansion and that's basically what I wrote here a point in generalized State space corresponds to a path that's exactly what I just said uh in virtue of this Taylor expansion here so a consequence of that is that if you look at the surprise of generalized State the negative  $\log P$  of vector  $X$  here and we call it the generalized LR gen it plays the role of an action scoring the likelihood of a local path over the correlation length of fluctuations so let me break down a bit this idea once again here the the the order of generalized motion depends on how smooth the noise is so basically when you you do this Taylor expansion um the um the time scale considered here uh corresponds to the the correlation length of your uh fluctuations the more uh correlated your noise is the more dynamical orders you have in a way the larger the path encoded by generalized States and the corresponding surprise the corresponding generalized Iran negative  $\log P$  of vector  $X$  is a quantity that score the likelihood of this path in virtue of the correspondence between points in generaliz state space and paths so a la a last remark I want to to do in order to be sure that uh everything is clear here is the following you might be a bit confused especially if you have a physics background uh because lents and actions are two different things so let's take a let's take a a step back and ask I have a path uh trajectory in state space if you will and I want to compute uh a probability density associated with a path how should I do so the first thing to do is to discretize time so basically you discretize your L equation so for the people acquainted with stochastic calculus just notice that it means choosing your Alpha discretization either EO or satovi let's say and from your your discretized L equation you define an infin an infinite asmal propagator just like for path integrals in quantum mechanics an infin alator is just a transition density between one state to the next during a Time window  $DT$  then you multiply the infinite asmal propagators and then you go to the continuous time limits and so in a nutal that how you proceed to compute to write down a density associated with a path and if you do that for instance in the simple case of a  $v$  process where the noise is a go and white noise you end up with the density of a path being proportional to the exponential of minus something and this something we call it the action so the smaller the action the more likely the corresponding path and this action in fact corresponds to an integral over your path of something and we call this something a Iran so a

the Iran is defined at any point in time while the action characterize the path as a wall thing so conceptually speaking Iren and actions are two very different things but here the key move is to uh kind of Leverage the idea that a point in generalized State space encode a path so that the surprise over generalized States this negative  $P$  of  $X$  which is defined as the generalized LR gen plays a role indeed of an action once again scoring the correlation length of fluctuations so that's really the core idea and Ling the path base formulation of the free energy principle so basically uh now that we have um introduced these ideas let's look at the generalized Iran over autonomous States so I just want to point out that throughout the presentation in virtue of this correspondence between points in generalized States and path I will um I will call generalized States path for instance here I would say the Iran of autonomous path I use these words in a interchangeable way and similarly when I I'm dealing with with autonomous path minimizing a generalized Iran I will be talking of uh the the path of least action because Iran play the role of an action so be aware of this terminology so if I'm looking at this generalized Iran of autonomous path so negative  $\log P$  of  $\alpha$  the cool thing here is that if I remove fluctuations noise on particular state or path so basically the particle respond deterministically to its environment I can rewrite this Iran like that and this formula coincides with an expected free energy so what this formula is all about well we can uh reorganize it and if we do so you can for instance write it that way so what it means is basically that uh the path minimizing this Iran so the most uh likely autonomous path average over all possible sensory path has to minimize these two terms and these two terms are very interesting because the first one if you look at it is it is an expected Iran over sensory path so minimizing this Iran means this expected Iran means following an autonomous path which yields unsurprising or preferred sensory path and and therefore it can be viewed indeed as an expected cost you want to minimize and if you minimize also the the second term this negative expected Information Gain you can see that it is just an expected K Divergence between two density and these two densities the difference between these two densities is just that the first one here is conditioned upon  $s$  so basically these two densities are different if the sensory path is informative so maximizing this expected Information Gain means following an autonomous path yielding uh an informative sensory path for instance let's say you are in a dark room where there is an ambiguous mapping between the hidden latent the hidden external States and the sensory stimuli let's say you can turn on the light and and this action would yield informative uh sensory inputs so that's the idea underlying the maximization of this expected Information Gain so this quantity is very rich and in fact you can do connections and links with many established ideas for instance you

can speak of um optimal basian decisions and optimal basian design or of uh pragmatic value and epistemic value Etc the whole idea here is that this quantity entails the preference seeking imperatives of the particles that the first term and the information seeking imperatives of the par particles that's the second term so the path minimizing this lound kind of constitutes the best direction of travel the optimal direction of travel for the particle so that you can view the particle as constantly engaging in an optimal Behavior okay so now having said that I want to go back to the world idea of synchronization map we talked about uh last week so in fact in this setting everything is exactly the same we just augmented State space if you will but you have  $\mu$  parameterizing a density everything is the same and if we consider the internal path of list action so the the internal path minimizing this Iran here the corresonding parameterized density over external path coincide with the density of external path given sensory path so that if you write down the corresponding free energy you have the first term which vanishes in virtue of this above equality here and free energy reduces to surprise if you will to this Iran over particular path I'm going to so before using this to uh derive directly the fery principle I need to uh briefly introduce uh the notion of generalized filtering so in a nutshell let's say that you have some data you have the vector  $s$  here a sensory path let's say and you want to compute negative  $\log P$  of  $s$  that's the quantity you ultimately want to evaluate but it's intractable let's say that Computing this  $P$  ofs would require a monstrous marginalization and you can't directly uh compute this guy instead you Define a proxy you define the an upper bound this variational free energy which is parameterized by  $\mu$  by Vector  $\mu$  and you just want to minimize this free energy so that it coincides with what you ultimately want to compute namely once again negative  $\log P$  of  $s$  so in order to minimize indeed this free energy you just follow uh a recognition or filtering Dynamics which is in fact a gradient sent on free energy in fact the the exact Dynamics at play here is this equation here so you could ask but why don't we just have the gradient term here  $\nabla F$  why do do we have a  $s$  term here this  $D\mu$  by the way about this  $D\mu$  here what is it well the Matrix  $D$  here is just this Matrix with ones here in in this uh line so basic basically if you have a vector  $X$  let's say  $x$   $x$  Prime  $X$  prime prime and so on and you want um to to to get from this Vector a shifted Vector  $X$  Prime the motion if you will you just have to apply this Matrix  $D$  here and on  $X$  and it gives you indeed  $X$  Prime and so basically you can view this  $D\mu$  here as being equal to  $\mu$  Prime basically and the idea is that you when once you the gradient term here is minimized when free energy is minimized so we we look at the stationary solution of this equation if you will  $m$  dot coincide with  $d\mu$  and let me just explain why we don't just have the gradient term here if

we only had the gradient term without the DM here we would go down on free energy until a minimum is reached and basically we would have the first component of the vector  $\mu$  being non zero but all the higher dynamical orders would be zero because we are at the minimum of free energy and we don't move anymore but here it is a wall Vector  $\mu$  which parameterize free energy and we don't want all the entries to be zero and we want an equation which guarantees that so that we add an additional term in the equation so that's basically in a natural what generalized filtering is all about and by the way note that if you consider the path of lift action of uh some process  $X$  you can write it in a very similar fashion the only difference is that here the quantity minimize is just the the this  $L$  generalized Iran here whereas it was free energy in the context of generalized filtering and by the way we just saw before that if uh we that if we consider the path the internal path of fist action the corresponding free energy here associated with the density the internal path of f action parametri it reduces to the Iran so that when we write down the equations here uh verified by the autonomous path of least action you can directly identify free energy to the Iran and we basically have the same equations that in generalized filtering so basically um if we look at the autonomous path of list action you can uh interpret the Dynamics of of the internal path of list action so in terms of basian inference which takes the form here of a basian filtering scheme and note that if we remove fluctuations on particular States or paths that the definition of dealing with a conservative particle the autonomous path will coincide with the autonomous path of least action and note that ignoring fluctuations might be relevant when it comes to very large particles uh where you can um so to very large particle you can cause grain and you can uh check these papers here where where they try to average out fluctuations using a renormalization group approach so in the end we find the free energy principle just as before uh but the difference is that here the the the the beian inference takes the form of a beian filtering scheme but otherwise we still end up with a variational principle accounting for the the for the perceptual and active inference that the particle do so okay now I want to do a bit of zology of typology let's say I said last time that this pass coping architecture here was a canonical Spar coping architecture but it was not a definitive feature of the free energy principle and that we could look at other Spar coupling architecture and I want now to look at various Spar coupling architecture so the most simple ones you could imagine is one like that where you don't have active States so basically um the the you only have the mark of blocket is only made of sensory state in the end you can still uh uh write down the internal path of list action like that but it will the idea that the particle en Cod believes about its external environment it will not manifest to an observer because by

definition an observer has only access to the mark of blanket and that's what this quote is all about when it says that whether internal Paths of Least action parameterized beliefs about external paths and therefore minimize variational free energy can only manifest via active states that is in active particles and of course no one would attribute any form of agency or sentience to a simple piece of rock for instance so if we move to the realm of active particles now we basically have the equations we uh studied before but the difference here with this pass coupling architecture is that the active states do not directly interact with the external States and the sensory states do not directly interact with the internal States this uh sparse coupling architecture is quite interesting because it reminds I would say it reminds the the way a bacteria is coupled with its environment because you would have like the outer M brain with all the transmembrane proteins The receptors whatever which would correspond to the sensory States and then you would have the underlying cortex filaments which mediate Med many aspects of biotic actions so I like I like very much this uh architecture here but we can even go further in the sophistication and consider the so-called strange particles the only move here is that we don't have any Arrow from  $s$  to  $I$  but more importantly we don't have any Arrow from  $a$  to  $\mu$  so the active states do not directly uh influence the internal States and this is interesting because it means that now the internal States alone are independent of both the active and the external States when conditioned upon the sensory state so so from the point of view of the internal States the active States become a latent cause just like the external states of the sensory of the sensory States so basically everything is the same you can write down a variational density parameterized by the internal States or path but now it is over both the external and the active path and so we have this free energy here we call it the generalized free energy uh  $G$  but everything is the same it's just the variational free energy associated with our recognition density which is over both external and active path so that in the end we have this equation here where you have in the equation of the internal path of least action the gradient on  $\mu$  on  $G$  here so it's very interesting because it basically means that the path infers infers and in fact cause uh the its own action  $\mu$  the the its own action which then cause its sensory uh inputs so and I'm quoting here the paper which introduced the notion of strange particles you can view the active particle the active path sorry as realizing the sensory consequences of the inferred action so basically such a particle kind of author its own action its own uh action and so the idea here is that the particle kind of infer its its own course of action which will yield preferred sensory outcomes so it's really a form of planning of inference strange particles do planning as inference so that's that's uh quite cool and in fact we can even go a bit further

and assume a certain level of sparsity within internal States so you would have this  $\mu_1$  here which influence  $\mu_2$  but not the other way around so from the point of view of the world all internal States everything is the same you have it parameterized all together density  $Q_{\mu}$  and by the way under a meanfield approximation you can write it that way where you would have  $q_{\mu_1}$  and  $Q_{\mu_2}$  here but the very cool thing here is that from the point of view of  $\mu_1$  alone well  $\mu_1$  is independent of  $\mu_2$  and ITA when condition upon  $s$  and you can view  $\mu_1$  alone as parameterizing a density over  $\mu_2$  and EA and we call it that way with the over script  $M$  so basically  $Q_{\mu_1}$  is a density or a belief about  $\mu_2$  and therefore about  $Q_{\mu_2}$  so that you can view this  $q_{\mu_1}$  as a metacognitive belief because it constitute a belief about a belief because it's a belief about  $Q_{\mu_2}$  so that just by assuming this simple architecture within internal States we end up with a um minimal beian mechanics let's say of metacognition and it also uh introduces the notion of metacognitive particles so I think that uh what ban mechanics is all about meaning uh translating cognitive abilities in simple physical term it constitutes in a way a physics of cognition if you will so H having said that uh thank you very much and especially thanks to all these guys here which who uh helped me a lot uh understanding uh the free energy principle and especially the path based formulation of the free energy principle thank you all right awesome great a lot of ways I could start but it's so interesting that um how smoothly the path-based continues on so maybe could you give a remark on just the timeline in the literature which one of these State and path base were developed just roughly like in which ordering was one first or how did happen yeah yeah so I'm not an expert on the history of the free energy principle but I know that thinking in terms of uh path and in terms of generalized filtering has been around for um forever in fact in the literature but about um the most um I would say uh definitive formulation in such terms I would say that the most important paper is um well I have it here uh it's called path integrals well where it is uh I can see it um well uh firston physics of Life review 2023b particular yeah exactly this paper here path integrals particular kinds and strange things which is also the paper so we which introduced um all this typology of particles and it kind of really uh formulated the frery principle that way um so I think that's pretty much a very very important paper and actually the one which introduce the notion of metacognitive particles the one of uh length and Lars um this one towards the bean mechanics of metacognitive particles the last one it is actually a commentary a short commentary very easily readable of the path integral paper we just mentioned um and also I would say that in the 2022 paper the fre energy principle made seem simpler but not too simple uh published in 2023 actually uh it is mainly it



mainly focused on the state based formulation but it it also talk about um paths and generalized States um so it it's also a very nice um very nice papers and it kind of deriv the whole thing from scratch so I think this paper is also very very important um and yeah I think basically in the last two or three years there have been many papers uh reformulated ating the free energy principle or grounding it in in in in solid mat and so on so it's I think currently and let's say the the few years which followed from the 2019 monography of car friston it has been very important years about when it comes specifically to beijan mechanics and the actual physics underlying the fire energy principle thank you yeah I'll just restate that I think there's a few great points to explore so even as early as the early 2000s there was the notion that there was like physics of Consciousness physics of cognitive systems a free energy principle for the brain there were physics-based equations in the 2010 friston paper there's a big tree with all these different inference algorithms however they were all just kind of branches on that tree little bit more evocatively or or aspirationally but not not formally then dot dot dot 2019 free energy principle for particular physics and the reading group and all the work around that and then especially in the last year since then um I think that the the decision and the move that you made to lead with the state based and then almost nothing had to be said today of course it was a great presentation but you said it all that we potify the path we make paths into points in a given space um and so I was thinking about like being in a car and then there's one path where like my first derivative is one for two seconds and then it stops and then that was that path and there's another path where my first derivative is one for 4 seconds and then I stopped and so all of those different paths which do take take the car to different locations they are also just points in this generalized space and then there's this interpretation of those coefficients as the Taylor series expansion but it just shows how versatile the formalism is because it can take points in arbitrary State spaces which means yes you can imagine arbitrary State spaces that correspond to Taylor series expansions or potentially other kinds of constructs what do we get from paths so in fact I think there are um a couple of things which uh motivated such a formulation because when I mean about the usual uh State based formulation we talked about last time there was um many critics many I mean it was there was many debates for instance about the question about the nest density being at steady state is it really relevant when it comes to real system to real biological systems for instance so there was many uh things which are now um addressed by the path based formulate formulation uh because as we saw we relaxed a couple of hypothesis and also the second thing which is very very much at the core of the path based formulation as you said is I mean the the the fact to move from um

states to path to path which are in fact encoded by generalized States it allows you to to to speak of the future uh to the to speaks to the Future Path so that you can develop a physics of particles planning for instance which was not at all the case with the previous formulation so having such extension of the scope of the free energy principle it allows you to to literally describe the basian mechanics of particles planning um and and so it's the fact to think in terms of path as opposed to States it allows you to develop a physics of let's say higher other cognitive abilities I would say uh so it's it's definitely very very cool and uh a great achievements let's say responding to that um I would say planning is possible in the state-based formalism it's just more of a Brute Force tree branching engineering problem so it's almost like in the state base formalism there was a physics of the perception action Loop and then there were classical computer science ways to deal with the branching of planning just like a chess algorithm like how many depth deep in the time Horizon and then there's all these secondary strategies for branching uh and pruning that surch tree but there's kind of like few to none in terms of the guarantee of the um time Horizons of policy you just kind of had to enumerate all the possible options um but the realtime colonel was physically grounded and beautiful and then planning had to be a little bit enumerated but when we have the path as a atomic entity um then we can kind of extend the the Elegance or the Simplicity that we were dealing with states in the moment but now our state in the moment is like a tailor expansion and this comes up a lot in the distinction between the discrete time and the continuous time models like figure 4.3 in the textbook where in a discreet time model if you want to plan 100 time steps in the future there's some variable you know  $T_{sub 100}$  like you literally are making a prediction and but you have no prediction for  $T_{sub 99.5}$  you're just making discrete predictions whereas a Taylor series even a tailor series for a super complex function and you're only going to go two levels of differentiation in you will have a prediction even for any point it might be radically wrong but you get the whole support from negative to positive Infinity basically for free without guarantees of it being accurate but it's like you've kind of pinned yourself to the timeline and then every single derivative that you take is giving you a better handle on that path for sure you'll never do worse and so that is like so similar yet also very different setting it's true that I said that the time scale scale considered was basically the correlation length of fluctuations on the number of the order of generalized motion but in principle your tailor expansion apply to I mean it's infinite but of course above beyond the correlation length of fluctuations it becomes wrong and I also want to say about what you said said in the beginning uh about the state based the the state based formulation yeah it's it's true

and actually in um in the paper the free energy principle simpler but not too simple they um they uh so you can write down the the action of a path so here we we have a Goan white no everything is is um so we are really in the state based formulation of the FP you can write the action of a path and you realize that it coincides with an expected free energy as well so you don't need to be in the path base formulation to uh go to to get to the expected free energy thing um which equates the action of autonomous path or here in our path base formulation it equates a generalized Iran of uh autonomous path so so yeah could you come back to where there was the Iran so this is um okay maybe uh one before this yes okay thank you for those of us outside of the nonactive physics world how is action used what does Action correspond to in physical systems when we're talking about action as generalized Iran is this the same thing as what we're talking about with policy selection and movement and embodied action how is this physics concept of action being used so first of all I would say that there is I mean it depends on what we're talking about um when it comes to actions in Iren we we usually think of um anal analytical mechanics but here it would be more in the context of stochastic calculus and in the context of stochastic calculus and path integrals in stochastic calculus the action is just a quantity which scores the likelihood of a path so if for instance the density of a path is uh the exponential of minus something you would call by identification this uh thing the action and basically the path because the higher the density of your path and vice versa so by the way when we we're saying for instance here that the action is equal to I mean we Define it as negative log density of a path in principle if the density I mean when it comes to the usual definition of the of the action if I write my density as a normalization factor times the exponential of minus the action then if I take the negative log of this density I would have the regular action plus something and usually this uh constant is dis is disgarded we um we just don't consider it so that the action reduces to negative log density of a path but conceptually speaking this action is very different from the idea of Iran because basically the ACT I mean usually speaking the action is the integral over time over your path of um of a quantity called the Iran so at every point in time there is a quantity defined and this is the Iran which is defined at any point in time while the action is a quantity which characterize path as W thing so conceptually speaking actions and lents are very different things Iran once again are defined at any point in time while the action okay LR was defined at any point in time while the action okay all wait a few seconds for Richard to rejoin if you're watching live please feel free to write questions in the chat and we'll look at them did you're back it's all good yeah Iran is defined at every point in time and then the action and basically um so in order to just uh

answer what you well uh here yeah when you said that is it different from the notion of action when in the active literature for instance and so on here we we use the word action uh but it has nothing to do with the notion of action like acting in the world Etc so yeah for the people who never um meet this concept so it can be super confusing I guess when you uh when you when you read page and mechanics papers you you don't really know what action is a we're talking about but yeah they are very different I know it's funny like each path has an action value the negative log density that in a way summarizes what we could say are the actions that that path entails but yet the actions or the affordances that are taken on that path that are summarized by the physics action um so there's yeah um also yeah very um Cutting Edge with reviewing the typology of particles and the dasta and saned Smith metacognitive particle um that shows I think a few things one that often there are implicit Concepts and qualitative Concepts that are built up like long have people said that there is a continuity of modeling between rocks and societies for example popes and plaintiffs and Plankton and all these other funny things that that friston at all say but not until this paper did we see the simple the conservative and the strange and then with that Target article just with the dasta and sanvit Smith work they kind of take that in another level and like compose off to the side and now we can go into a metacognitive depth combining back to like the hierarchical nested meta awareness work of saned Smith from 2021 so it's just very cool how there's like a kind of earlier first pass qualitative intuition and then the Empirical research question is like where can we build the High-Speed Rail lines and this is like the blue print for the highspeed rail now and then the next level will be like actually making the simulations or whatever it is to kind of show that this is not just something that you can make in PowerPoint but this is actually something where the rabbit is going to be evincing some kind of behavior that it couldn't otherwise but it's kind of like that's like in a way it's not the only way but it is like an agenda between the um intuitive to the sketched and just builds forward in these ways that are being reviewed pretty clearly and changing on a month by month yeah and I think I mean this I I mean this agenda of translating um Concepts like cognitive abilities and stuff in simple physical terms is quite interesting and I I mean we we discussed it on by message on Discord but it's it's um it would be cool and it's it's coming it's happening but uh it's it could be cool to for the this uh ban mechanics this field to be more known and recognized by the physics Community because if you meet like a regular physicist working for instance in the physics of complex systems on or in biological physics or bioh physics and stuff like that it I mean you like a 0.999 probability that he never heard of basan mechanics while we have a world agenda to to to to to following

is I think there is way more to to be done it's like the the down of this uh field of this agenda let's say so I think the the next years are quite exciting in that regard I I totally agree like one meme or theme on that for me is like the base graph on the screen on the table it is what it is and then there's the second level where we annotate or assert that graph with cognitive phenomena like well this graph reflects attention or awareness of attention or metacognition or regret or whatever we're modeling we kind of make an assertion about the base graph and like you said it's the dawn this isn't the answer on metacognition this is like one very Snappy very um following in line way of modeling metacognition there's no end there's no end to that question of how do you model metacognition and so then it's just like every time in a psychology paper or any we see like a cognitive phenomena which which can include everything from anticipation you know past present future that's like our Keyhole to bridge with this generalized unifying perspective on cognitive systems yeah I totally agree this work is really um this work here on the slide is very like a first work but I mean when it comes to metacognition for instance I mean I'm not an expert at all but there are many Concepts I don't know m actions cognitive effort whatever there are many things many layers and the the I think like translating all of these concepts of these phenomena processes whatever into an actual basing mechanics is um is really like the what's coming in fact I think in the next years so and yeah we'll see and and as you said beyond the analytical work there is also the simulations work examples whatever um so yeah I think there are many many Works to to be done and which are coming in the in the next years I totally agree yeah kind of analogy that brings to mind is like early in the periodic table not saying that we're studying material phenomena um or even like elements are material phenomena but it's kind of like well the has Vision taste and hearing so if we've identified uh memory in vision and taste there's like a missing element for maybe it's not there maybe the memory for hearing is zero or maybe it's something very complex but it's like but there's a space there it's like there should be um a rare earth metal in the fourth row there should be an attention variable on this and so it's kind of like a higher dimensional periodic table where because we know about certain patterns that are either kind of convergently arising in the real world because of what is life or they're um convergently arising because we choose to model things a certain way then that brings a huge amount of concordance and juxtaposition like to the field of chemistry whereas previously there might have been air chemistry and water chemistry and fire chemistry and then there'd be like oh but what happens when you throw water on a fire or what happens when the air in the room Burns Out there'd be like these edge cases where it's like oh we don't do that but

that's what's happening when we don't have the unified model for cognitive systems people will kind of study one phenomena or character of a system to its limits but the limits of any phenomena in our highly woven life forms like you don't chase it to the end if you're studying foraging in the ant colony it's not like there's a okay that was the end of the rainbow foraging is over it's like oh well foraging is related to nursing and then that's related to this and the weather it it never just simply terminates with the inquiry and um um possibly that could reflect the extremely early stage of formalization possibly there are fundamental unknowns and adjacencies in our epistemic situation yeah and I think I mean few years ago the idea of kind of extending physics to in order to have an actual physics of cognition it would be like uh crazy like the holy grail and the fact that we have this early works is quite an achievement and I would also say that what's interesting here is that we already do have um formal uh models of many um of many cognitive um phenomena for instance you were talking about the the paper of last about metacognition um so we have very precise and formal active inference models for instance of many different phenomena like metacognition and so on so in a way we kind of already know what we should end up with and the idea is uh how to do how to play for instance with the sparsity within internal States in order to go to get back or to rederive or refind what people are already modeled in a different literature earlier so that's kind of the Dynamics I think um so yeah we'll see in the next year how it goes yeah that's a great point a few more thoughts on that like a lot of the uh adjacencies and generalizations mathematicians and physicists and statisticians they're experts in this like if it was a fixed number see if it could be variable if there wasn't a variance see if you can add a variance if it was this kind of distribution swap it out for like there are these kind of syntactic moves that barely require much more than just like yeah these are the swaps we make like this is what it looks like to generalize a theory it's like we could add a variance on you know all all so those kinds of um familiar moves and then um so much can be explore on this topic but when we think about a physics of cognitive systems or a physics of cognition sometimes I feel like I have one foot like is it's like tied or somehow trailing with the reductionism like it's hard to escape the material basis and it's also not clear if we want to escape the material basis because of physics of cognitive systems on one hand it might address this kind of like Thermo info neant Tropic schinger what is life question about persistence and Order of kind of Quasi crystals or leave all that mess separate mess and Beauty the pure basian mechanics if we just say okay now we're in the map we're we're leaving the territory behind we're not going to talk about the actual calories and the thermodynamics we're just on the map and imag massive prior with a little

attention is like a piece of sand hitting a mountain and a very loose prior with a highly attended to data point is like a bowling ball you know smashing through a small little um heap of sand so it's like there's a physics to the Collision of incoming data and prior which is the basian setting so it's interesting that there's a sort of map basian mechanics that kind of noncontroversial describes the Collision of priors of different mass in a way and then the tantalizing question or connection is whether that is like one in the same or an enabling factor or a downstream factor of this actual informational autopoesis for living systems if I may add something I think here the uh I mean an interesting point is that if you have for instance um I mean I I don't want to go there really but if I we have if I have a software which is which can be implemented by many sort of Hardwares for inance um and this idea was actually a lot debated in the when it came to functionalism and stuff like that but anyway here the such basian mechanics in a way it's the physics of um how um how the the overall system should behave but we so there is for instance the minimization of this uh free energy functional but we are not saying how the here we we do not um uh explicitly say how the system uh actually does the computations we just say that a sply coer romal system has somehow to do that stuff and in a way it is neutral about what um what the the actual system is made of um it's like when we say that the fery principle is more a framework than a process Theory or stuff like that um but I mean I I I um I don't know very much about these uh questions I I I think that's that's a great point it's like that's why a program or an operating system can be run when a different processor is brought in just within a kind of classical Computing setting um as a cognitive mapping or cognitive modeling framework like we don't know if that rabbit is a real biological rabbit that rabbit might be a deep fake rabbit and we're interacting with it through a video channel and we think that we're doing behavioral research on a real rabbit but it's just a really convincing synthetic data rabbit um we don't no because we're only getting the observations that we're getting and then we can tell any number of possibly compatible stories or interpretations about the data we're getting and to say more or to go further like that's stepping past the boundary of the actual observations we're making which is great like we want to propose hypotheses for what we're not directly observing that's the whole point of latent States and everything and yet there is this line where it's like you can kind of falsely push the known knowns Beyond where they really are um and it seems like one example of that is going down and specifying the actual mechanistic basis of a given computational function and that that's if that's to specify then the work itself is to specify it however at the framework level the framework's absence of that kind of a material substrate

like that's the feature that's not a A lacina in the framework that is like there's the USB stick and then there's where you can plug it in that is the plugin to any system and if there was something something already pre- plugged in there like well it has to happen on Turing computer or it can't happen on Turing computer that would have just hobbled the scope of applicability by welding together what doesn't need to be welded and I mean if anything this is a um journey and a a challenge about proper articulation and about how sparsity and nuancing which are connected to what brings us cognitive phenomena so like many things to learn and reflect on yeah and if I may add a a related remark um about the the question of having a topdown approach as as opposed to bottom of approach I think sociologically speaking and historically speaking it's very interesting that the people who started developing basan mechanics are people who are not like supposed to be professional physicist because I think when you it comes to like super emergent kind of phenomena like cognition whatever um the the people who was training is precisely to learn uh about cognition about agencies the people who kind of know everything about what it takes to be an agent and so on these guys are neuroscientist basically so they because they know what agency entails they can be the good ones to propose an actual ban mechanics which is a very uh top down approach whereas if you go to the labs in like a physics of complex systems lab and stuff like that people are pretty much um bottom up they will for instance oh I I want to study the brain so I'm going to write down the upfield model for instance so a kind of izing model where up is an activated neurons and down um uh uh uh non not activated neurons and so I will have this very this very much bottom up approach and I can which is super interesting and super important but when it comes to this very much uh emergent Behavior Uh the the very emergent phenomena of the brain metacognition blah blah blah I can study for like five centuries the upfield model I will never get any insight about the emerging behavior of the brain and um and um so I think here you have to have a a top down approach you have to write a very generic uh L equation and because you're a neuroscientist and because you know what it takes to do an inference to to be an inference machine you're like ah conditional independencies is very different is very important sorry so let's try to inject in my L equation some level of spse coupling and then boom you have the free energy principle so even So sociologically speaking What's Happening Here might be super weird like these guys are not supposed originally at least to be physicist and they are supposed to like kind of generate the next chapter of physics how weird is that but in fact in fact I think they were precisely the the good people to do that um so yeah it's it's quite interesting I think there might be oh that's a great point there might be a fun



history with physician physician heal thyself and the alliance of medicine and physic that's one point another point is um low road and high road and sometimes the the disciplinary or the ingroup conversation is like very low road and then the idealistic and aspirational is like High Road it doesn't exactly map to those but I'm thinking of like somebody says like I want to drive somewhere I want to travel somewhere there is no material basis for that travel yet they're not there there's no path there's no car and then in the mechanic shop it's like well we have this tool this object like the low road we're so surrounded by the low road that it supports this incremental research agenda using the tools and approaches that we have and their materiality whereas someone from outside field um like comes to the ant researcher it's like have you had the ants build this new thing and it's like well we we weren't even on that path but now that's like a new North Star and that can now draw work in that direction and so it's kind of like the the low road building out and then like the kind of um the draw of the adjacent possible in the imagination that the high road kind of grounds in all right I'll ask a question from the live chat Susan asks what can one equation contain I'm imagining how to interpret and translate self-modeling how to approach foraging resources and opportunities uh can you can you repeat please yes it's an open-ended question so yeah feel free to however you like what can one equation contain I'm opportunities so I I I I could not answer the second part of the question but if I can pick up on uh about the question of uh what what these equations are all about which is a bit what you asked last week like what Works do does the framework do what these equations contain or whatever um if you actually simulate a sparsely couple random dynamical system and actually there are there already are examples out there for instance in the paper called uh uh Mar stochastic chaos and Mark of blankets if I remember well with Thomas for and car fron and others they simulate a a coupled uh low rent um uh systems if I if I remember well and if you do um simulate such systems which verifies sear SP coping architecture You observe the Behavior which is described by your equations and if for instance you remove fluctuations on particular States so uh well your simulation which follows the path of list action which is given by your equation so it's it's um it's it's uh it does really work in the sense that it does describe indeed something it's really a physics in fact of such uh SP systems great question I think we could Explore More on what an equation can contain but I'm just kind of struck by that there's like a latent large set of um to be clarified axioms and conclusions and so on and when we see like one equation written here it's kind of like we're just seeing like one glimpse or one facet it's not like the other equations aren't in effect but sometimes it's a little unclear which equations are relevant but in an equation um we see a

symbolic expression that within a given Quantum reference frame other otherwise the Q and the a wouldn't mean anything like with within a semantic reference frame the equation is just kind of like a check it's like if it's like it it's there's probably many ways that we can kind of explore what it does but it's true that I mean I think many peoples who in addition don't necessarily have uh mathematical background or whatever they have been I I I mean I'm not an expert in the history of the fery principles but a lot of people have been confused especially in the early steps where it was as I say it many times where it when it was even more an intuition than a solid framework properly grounding it solid mat so people did get a bit confused I think by with the mat but now it's we have more and more worked examples simulations um solid mat Etc so it's it's quite um quite nice okay in closing we have explain it like I'm a 10-year-old now that that response may reveal more about what somebody thinks 10-year-olds are like but now that we've been on this two-part Journey you know dotzip it and close it Richard so explaining you mean uh about what I just said like uh before just both sections or just our whole projects now that now that we can look back and all the incredible quality summaries that we've provided how do we just cap it and move forward with the child yeah so I think everything we did here so both in this presentation and the previous one is um so basically we can so if I I I would so in order to take in the most um intuitive and simple ways I would say that very simply put a system is just um it's just um it's just some things interacting let's say so it would literally correspond to a system as we use this word in a in a daily basis uh uh pen is a system an organism or any piece of uh any collection of things interacting with each other we I'm I can say this is the system I just have to precisely delimitate um and and say what are the boundary of my system and given a system which has a certain Dynamics um certain laws physical laws telling you how your system behave how the things move around and so on and now so my system can be alone like an asteroid in space let's say or it can be connected to each to other things I recognize as other systems um two things can Collide two different things systems can collide with each other etc for instance and now if I assume several systems which are connected meaning that they Cally interact with each other in a certain SP way so basically in my schematic you would have for instance a system on the right a system on the left and a sub system in the middle so you have many subsystems coal interacting in a sparely way for instance mu doesn't interact with ITA if I take a bacteria for instance it does move around and interact with other things outside the bacteria but at the same time within the bacteria under the outer membrane there there is also stuff which do not interact because that interact with the outside let's say and so if you have such uh

sparse coupling architecture as we say we and were and in in in a more technical term terms we would say that the inside is uh independent of the outside when conditioned on the intermediate system let's say so if we have such sply coupled system dynamical system uh we see mathematically speaking we derive the fact that your system has to behave in a certain way and in fact when you do have such spy coule dynamical system you realize that everything uh appears to uh I mean if you look at for instance internal States  $\mu$  here the Dynamics of system uh can be interpreted in terms of beian inference meaning that this system can be viewed as encoding a density over its external States for instance and if I had to Define in a very simple way what it means to parameterize or to encode a distribution I would say that um your um the various uh parameters which Define your system for instance uh I said earlier that a system could be anything a pen an asteroid whatever there are variables which specify its St for instance temperature pressure shape whatever I have a set of variables and I can view these variables these numbers literally as I can take them and manipulate them as parameters of a mathematical quantity I I I can deal with and yeah so I'm not sure if I failed in trying to put it in in a very very simply terms but I I hope it makes sense it was a awesome Richard thank you for um working to bring us these twin gems in the distillation work and really helping like consolidate and sediment on one hand a long history of dynamical systems and physics and on the other hand papers coming out two weeks ago as well so hopefully we'll continue the adventure and um see you around yeah thank you and thank you for the chat questions on the chat cool peace everyone bye bye thank you I